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Experience

BlueStock – SDE Intern *Apr 2025 – May 2025*

Software Development Engineer Intern Remote

- Led a **10-member team** to build a full-stack IPO platform using **Node.js**, **Express**, **MongoDB**, **React**, and **Tailwind CSS**, featuring **30+ IPO listings**, live CMP updates, listing gains tracking, and RHP/DRHP PDF access.
- Designed and implemented **10+ secure RESTful APIs** with **JWT authentication**, Mongoose schemas, and **Multer + AWS S3** integration to handle **100+ company assets**.
- Streamlined Agile workflows using **GitHub** and **Postman**, accelerating feature delivery speed by **25%** while managing admin dashboards and database structuring.

IIT Patna – Machine Learning Research Intern *Oct 2023 – Jan 2024*

Satellite Imagery Segmentation & Deep Learning Pipeline Patna, Bihar

- Orchestrated an end-to-end **Computer Vision** pipeline using **PyTorch** and an **Attention-Gated Residual U-Net (R2U-Net)**, training on **1.2M+ parameters** for flood-water semantic segmentation across **500+ multispectral patches**.
- Curated and augmented **100+ high-resolution satellite rasters** using **Label Studio** and **OpenCV**, boosting segmentation precision and achieving a **15% higher Mean IoU / Dice coefficient** on test validation.
- Optimized data preprocessing loaders and environment runtime configurations via **Anaconda** and **CUDA**, reducing training pipeline latency by **40%** across **50+ test iterations**.

Projects

Portfolio

Responsive developer portfolio built with React (Vite) and Tailwind CSS.

- Engineered **10+ modular components** for cross-device compatibility and published the web application on **Vercel**.
- Embedded **particle animations**, **lazy loading**, smooth scrolling, and scroll-triggered transitions to enhance UI performance and page responsiveness.

Aria – Full-Stack AI Assistant & Agentic Workflow

Autonomous browser-based AI assistant with persistent memory and multimodal tool orchestration.

- Architected an agentic AI assistant using **LangGraph**, **LangChain**, and **Flask**, orchestrating **5+ tools** for web search, arithmetic, and multimodal parsing across **4 file formats (PDF, DOCX, CSV, TXT)** via **Ollama Cloud API**.
- Encoded stateful conversation persistence using **PostgreSQL** checkpoints on **Supabase**, integrating speech-to-text, audio synthesis, and a **5 MB file pipeline** hosted with **Gunicorn** on **Render**.

Street Light Complaint Management System (SLCS)

Role-based civic complaint management platform.

- Developed a **MERN-stack** complaint lifecycle platform supporting secure authentication and real-time status tracking for **100+ simulated users**; deployed on **Render**.
- Pioneered **10+ RESTful APIs** featuring **JWT authentication**, **bcrypt hashing**, role-based access control (**RBAC**), and multipart uploads with **Multer** and **MongoDB**.

Education

National Institute of Technology Agartala *2022 – 2026*

B.Tech in Electrical Engineering CGPA: **7.57/10**

Tulsi Vidya Niketan, Varanasi *2022*

CBSE Class XII Percentage: **91.8%**

Tulsi Vidya Niketan, Varanasi *2020*

CBSE Class X Percentage: **93.2%**

Technical skills

Languages: C++, JavaScript, Python, HTML/CSS

AI Tools: GenAI, LangChain, LangGraph, RAG, Agentic AI, AI Agents

Frameworks & Tools: Node.js, Express.js, React, MongoDB, Firebase, Git, GitHub

Core CS Fundamentals: Object-Oriented Programming, Operating Systems, DBMS, Computer Networks

Coding profiles

- LeetCode: Solved **1000+ DSA problems**, max rating: **1807** (handle: dipcoderr).
- CodeChef: **3-star coder**; max rating: **1611** (handle: dipcoderr).